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Lab Assignment 1

UNIX Commands

AIM: To study and execute the commands in Unix.

COMMAND:

1.Date Command:

This command is used to display the current data and time.

Syntax:

$date

$date +%ch

Options: -

a = Abbrevated weekday.

A = Full weekday.

b = Abbrevated month.

B = Full month.

c = Current day and time.

C = Display the century as a decimal number.

d = Day of the month.

D = Day in 'mm/dd/yy' format

h = Abbrevated month day.

H = Display the hour.

L = Day of the year.

m = Month of the year.

M = Minute.

P = Display AM or PM

S = Seconds

T = HH:MM:SS format

u = Week of the year.

y = Display the year in 2 digit.

Y = Display the full year.

Z = Time zone .

To change the format :

Syntax :

$date '+%H-%M-%S'

```
ell@DESKTOP-LVMF4QJ:~$ date
Mon Feb  9 10:16:06 UTC 2026
ell@DESKTOP-LVMF4QJ:~$ date +"%H-%M-%S"
10-16-40
ell@DESKTOP-LVMF4QJ:~$
```

2.Calender Command:

This command is used to display the calendar of the year or the particular month of calendar year.

Syntax:

a.$cal <year>

b.$cal <month> <year>

Here the first syntax gives the entire calendar for given year & the second Syntax gives the calendar of reserved month of that year.

Common & Useful Variations

1. **Specific month & year:**
   cal 2 2026 -> shows February 2026
2. **Whole year view:**
   cal 2026 -> Shows all 12 months in a compact grid.

```
ell@DESKTOP-LVMF4QJ:~$ cal
   February 2026
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28


ell@DESKTOP-LVMF4QJ:~$ cal 7 2026
     July 2026
Su Mo Tu We Th Fr Sa
          1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31
```

3.Echo Command :

This command is used to print the arguments on the screen .

Syntax : $echo <text>

Multi line echo command :

To have the output in the same line , the following commands can be used.

Syntax : $echo <text\>text

To have the output in different line, the following command can be used.

Syntax : $echo “text

>line2

>line3”

```
ell@DESKTOP-LVMF4QJ:~$ name="Jaspreet"
ell@DESKTOP-LVMF4QJ:~$ echo $name
Jaspreet
ell@DESKTOP-LVMF4QJ:~$ echo "OS"
OS
ell@DESKTOP-LVMF4QJ:~$ echo "My Name is Jaspreet Kaur.I am from Amritsar"
My Name is Jaspreet Kaur.I am from Amritsar
ell@DESKTOP-LVMF4QJ:~$
```

4.Banner Command :

It is used to display the arguments in ‘#’ symbol .

Syntax : $banner <arguments>

```
ell@DESKTOP-LVMF4QJ:~$ banner Operating System
#######
#     #  #####   ######  #####     ##     #####     #    #    #   ####
#     #  #    #  #       #    #   #  #      #       #    ##   #  #    #
#     #  #    #  #####   #    #  #    #     #       #    # #  #  #
#     #  #####   #       #####   ######     #       #    #  # #  #  ###
#     #  #       #       #   #   #    #     #       #    #   ##  #    #
#######  #       ######  #    #  #    #     #       #    #    #   ####

 #####
#     #   #   #   ####    #####  ######  #    #
#          # #   #         #    #       ##  ##
 #####      #     ####      #    #####   # ## #
      #     #         #     #    #       #    #
#     #     #    #    #     #    #       #    #
 #####      #     ####      #    ######  #    #

ell@DESKTOP-LVMF4QJ:~$
```

5.’who’ Command :

It is used to display who are the users connected to our computer currently.

Syntax : $who – option’s

Options : -

H–Display the output with headers.

b–Display the last booting date or time or when the system was lastely rebooted.

```
ell@DESKTOP-LVMF4QJ:~$ who -H
NAME     LINE         TIME             C
ell      pts/1        2026-02-09 10:15
ell@DESKTOP-LVMF4QJ:~$ who -b
         system boot  2026-02-09 10:15
```

6.'who am i' Command :

Display the details of the current working directory.

Syntax : $who am i

```
ell@DESKTOP-LVMF4QJ:~$ whoami
ell
```

7.'tty' Command :

It will display the terminal name.

Syntax : $tty

```
ell@DESKTOP-LVMF4QJ:~$ tty
/dev/pts/0
```

8.'Binary' Calculator Command :

It will change the '$' mode and in the new mode, arithematic operations such as +,-

,*,/,%,n,sqrt(),length(),=, etc can be performed . This command is used to go to the binary

calculus mode.

Syntax :

$bc operations

^d

$

1 base –inputbase

0 base – outputbase are used for base conversions.

Base :

Decimal = 1 Binary = 2 Octal = 8 Hexa = 16

```
ell@DESKTOP-LVMF4QJ:~$ bc
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type `warranty'.
obase=6
4
1110
1110
234
234
obase=8
2
1100
1100
ibase=2
8
1100
1100
obase=2
10
1010
obase=8
1100
2114
```

9.'CLEAR' Command :

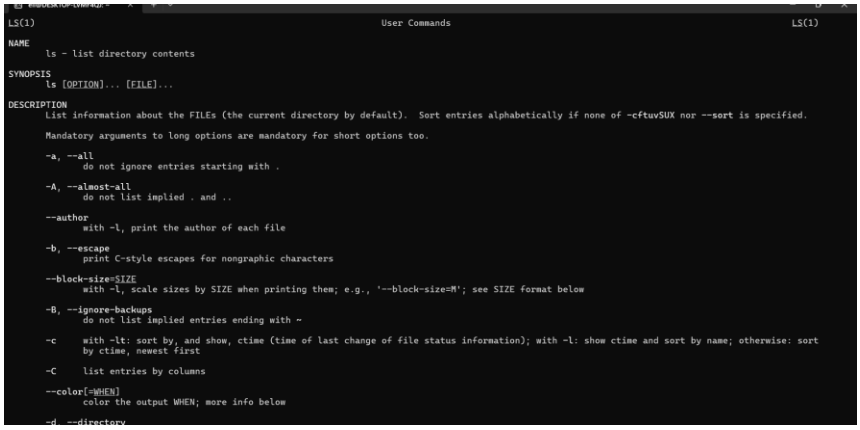
It is used to clear the screen.

Syntax : $clear

10.'MAN' Command :

It help us to know about the particular command and its options & working. It is like 'help' command in windows .

Syntax : $man <command name>



```
LS(1)                                User Commands                                LS(1)

NAME
       ls - list directory contents

SYNOPSIS
       ls [OPTION]... [FILE]...

DESCRIPTION
       List information about the FILEs (the current directory by default).  Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              do not ignore entries starting with .

       -A, --almost-all
              do not list implied . and ..

       --author
              with -l, print the author of each file

       -b, --escape
              print C-style escapes for nongraphic characters

       --block-size=SIZE
              with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below

       -B, --ignore-backups
              do not list implied entries ending with ~

       -c     with -lt: sort by, and show, ctime (time of last change of file status information); with -l: show ctime and sort by name; otherwise: sort
              by ctime, newest first

       -C     list entries by columns

       --color[=WHEN]
              color the output WHEN; more info below

       -d, --directory
```

11.MANIPULATION Command :

It is used to manipulate the screen.

Syntax : $tput <argument>

Arguments :

1.Clear – to clear the screen.

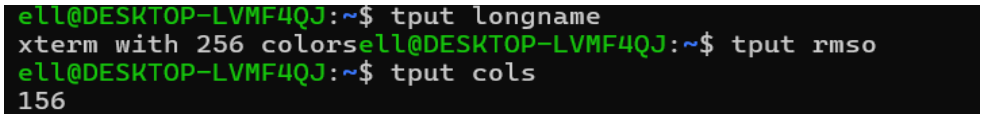
2.Longname – Display the complete name of the terminal.

3.SMSO – background become white and foreground become black color.

4.rmso – background become black and foreground becomes white color.

5.Cop R C – Move to the cursor position to the specified location.

6.Cols – Display the number of columns in our terminals.



```
ell@DESKTOP-LVMF4QJ:~$ tput longname
xterm with 256 colorsell@DESKTOP-LVMF4QJ:~$ tput rmso
ell@DESKTOP-LVMF4QJ:~$ tput cols
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```

12.LIST Command :

It is used to list all the contents in the current working directory.

Syntax : $ ls – options <arguments>

If the command does not contain any argument means it is working in the Current directory.

Options :

a– used to list all the files including the hidden files.

c– list all the files columnwise.

d- list all the directories.

m- list the files separated by commas.

p- list files include ‘/’ to all the directories.

r- list the files in reverse alphabetical order.

f- list the files based on the list modification date.

x-list in column wise sorted order.

```
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ell@DESKTOP-LVMF4QJ:~$ ls -a
.  ..  .bash_history  .bash_logout  .bashrc  .cache  .lesshst  .motd_shown  .profile  .sudo_as_admin_successful  Directory1  dev  file  test
ell@DESKTOP-LVMF4QJ:~$ ls -c
Directory1  dev  file  test
ell@DESKTOP-LVMF4QJ:~$ ls -d
.
ell@DESKTOP-LVMF4QJ:~$ ls -m
Directory1, dev, file, test
ell@DESKTOP-LVMF4QJ:~$ ls -p
Directory1/  dev/  file  test/
ell@DESKTOP-LVMF4QJ:~$ ls -r
test  file  dev  Directory1
ell@DESKTOP-LVMF4QJ:~$ ls f
ls: cannot access 'f': No such file or directory
ell@DESKTOP-LVMF4QJ:~$ ls -f
.bashrc  ..  .bash_history  .sudo_as_admin_successful  test  .bash_logout  .  .profile  dev  .motd_shown  .lesshst  .cache  Directory1  file
ell@DESKTOP-LVMF4QJ:~$ ls -x
Directory1  dev  file  test
ell@DESKTOP-LVMF4QJ:~$
```

DIRECTORY RELATED COMMANDS :

1.Present Working Directory Command :

To print the complete path of the current working directory.

Syntax : $pwd

2.MKDIR Command :

To create or make a new directory in a current directory .

Syntax : $mkdir <directory name>

3.CD Command :

To change or move the directory to the mentioned directory .

Syntax : $cd <directory name.

4.RMDIR Command :

To remove a directory in the current directory & not the current directory itself.

Syntax : $rmdir <directory name>

```
ell@DESKTOP-LVMF4QJ:~$ pwd
/home/ell
ell@DESKTOP-LVMF4QJ:~$ mkdir project
ell@DESKTOP-LVMF4QJ:~$ pwd
/home/ell
ell@DESKTOP-LVMF4QJ:~$ cd project
ell@DESKTOP-LVMF4QJ:~/project$ pwd
/home/ell/project
ell@DESKTOP-LVMF4QJ:~/project$
```

FILE RELATED COMMANDS :

1.CREATE A FILE :

To create a new file in the current directory we use CAT command.

Syntax :

$cat > <filename.

The > symbol is redirectory we use cat command.

2.DISPLAY A FILE :

To display the content of file mentioned we use CAT command without '>' operator.

Syntax :

$cat <filename.

Options –s = to neglect the warning /error message.

3.COPYING CONTENTS :

To copy the content of one file with another. If file doesnot exist, a new file is created and if the file exists with some data then it is overwritten.

Syntax :

$ cat <filename source> >> <destination filename>

$ cat <source filename> >> <destination filename> it is avoid overwriting.

Options : -

-n content of file with numbers included with blank lines.

Syntax :

$cat –n <filename>

4.SORTING A FILE :

To sort the contents in alphabetical order in reverse order.

Syntax :

$sort <filename >

Option : $ sort –r <filename>

5.COPYING CONTENTS FROM ONE FILE TO ANOTHER :

To copy the contents from source to destination file . so that both contents are same.

Syntax :

$cp <source filename> <destination filename>

$cp <source filename path > <destination filename path>

6.MOVE Command :

To completely move the contents from source file to destination file and to remove the source file.

Syntax :

$ mv <source filename> <destination filename>

7.REMOVE Command :

To permanently remove the file we use this command .

Syntax : $rm <filename>

```
ell@DESKTOP-LVMF4QJ:~/project$ cat > file.txt
Hello,This is Jaspeeet.
Student of NIT Jalandhar
ell@DESKTOP-LVMF4QJ:~/project$ cat file.txt
Hello,This is Jaspeeet.
Student of NIT Jalandhar
ell@DESKTOP-LVMF4QJ:~/project$ cat -s file.txt
Hello,This is Jaspeeet.
Student of NIT Jalandhar
ell@DESKTOP-LVMF4QJ:~/project$ cat -n file.txt
     1  Hello,This is Jaspeeet.
     2  Student of NIT Jalandhar
ell@DESKTOP-LVMF4QJ:~/project$ cat file.txt > dest.txt
ell@DESKTOP-LVMF4QJ:~/project$ cat file.txt >> destt.txt
ell@DESKTOP-LVMF4QJ:~/project$ cat dest.txt
Hello,This is Jaspeeet.
Student of NIT Jalandhar
ell@DESKTOP-LVMF4QJ:~/project$ cat destt.txt
Hello,This is Jaspeeet.
Student of NIT Jalandhar
ell@DESKTOP-LVMF4QJ:~/project$ sort file.txt
Hello,This is Jaspeeet.
Student of NIT Jalandhar
ell@DESKTOP-LVMF4QJ:~/project$ sort -r file.txt
Student of NIT Jalandhar
Hello,This is Jaspeeet.
ell@DESKTOP-LVMF4QJ:~/project$ cp file.txt des.txt
ell@DESKTOP-LVMF4QJ:~/project$ mv file.txt file1.txt
ell@DESKTOP-LVMF4QJ:~/project$ rm file1.txt
ell@DESKTOP-LVMF4QJ:~/project$ cat file1.txt
cat: file1.txt: No such file or directory
ell@DESKTOP-LVMF4QJ:~/project$
```

8.WORD Command :

To list the content count of no of lines , words, characters .

Syntax :

$wc<filename>

Options :

-c – to display no of characters.

-l – to display only the lines.

-w – to display the no of words.

```
ell@DESKTOP-LVMF4QJ:~/project$ wc destt.txt
 2  7 51 destt.txt
ell@DESKTOP-LVMF4QJ:~/project$ wc -c destt.txt
51 destt.txt
ell@DESKTOP-LVMF4QJ:~/project$ wc -l destt.txt
2 destt.txt
ell@DESKTOP-LVMF4QJ:~/project$ wc -w destt.txt
7 destt.txt
```

9.LINE PRINTER :

To print the line through the printer, we use lp command.

Syntax :

$lp <filename>

10.PAGE Command :

This command is used to display the contents of the file page wise & next page can be viewed by pressing the enter key.

Syntax :

$pg <filename>

```
ell@DESKTOP-LVMF4QJ:~/project$ lp destt.txt
lp: Error - No default destination.
ell@DESKTOP-LVMF4QJ:~/project$ pg file1.txt
pg: command not found
```

11. FILTERS AND PIPES

HEAD : It is used to display the top ten lines of file.

Syntax: $head<filename>

TAIL : This command is used to display the last ten lines of file.

Syntax: $tail<filename>

PAGE : This command shows the page by page a screenfull of information is displayed after which the page command displays a prompt and passes for the user to strike the enter key to continue scrolling.

Syntax: $ls –a\p

MORE : It also displays the file page by page .To continue scrolling with more command press the space bar key.

Syntax: $more<filename>

GREP : This command is used to search and print the specified patterns from the file.

Syntax: $grep [option] pattern <filename>

SORT : This command is used to sort the datas in some order.

Syntax: $sort<filename>

PIPE : It is a mechanism by which the output of one command can be channeled into the input of another command.

Syntax: $who | wc-l

TR : The tr filter is used to translate one set of characters from the standard inputs to another.

Syntax: $tr “[a-z]” “[A-Z]”

```
ell@DESKTOP-LVMF4QJ:~/project$ head destt.txt
Hello,This is Jaspeeet.
Student of NIT Jalandhar
ell@DESKTOP-LVMF4QJ:~/project$ tail destt.txt
Hello,This is Jaspeeet.
Student of NIT Jalandhar
ell@DESKTOP-LVMF4QJ:~/project$ more destt.txt
Hello,This is Jaspeeet.
Student of NIT Jalandhar
ell@DESKTOP-LVMF4QJ:~/project$ grep Jaspreet destt.txt
ell@DESKTOP-LVMF4QJ:~/project$ grep "Jaspreet" destt.txt
ell@DESKTOP-LVMF4QJ:~/project$ grep "is" destt.txt
Hello,This is Jaspeeet.
ell@DESKTOP-LVMF4QJ:~/project$ more destt.txt
Hello,This is Jaspeeet.
Student of NIT Jalandhar
ell@DESKTOP-LVMF4QJ:~/project$ sort destt.txt
Hello,This is Jaspeeet.
Student of NIT Jalandhar
ell@DESKTOP-LVMF4QJ:~/project$ sort -r destt.txt
Student of NIT Jalandhar
Hello,This is Jaspeeet.
ell@DESKTOP-LVMF4QJ:~/project$ who | wc -l
1
ell@DESKTOP-LVMF4QJ:~/project$ cat destt.txt | tr "[a-z]" "[A-Z]"
HELLO,THIS IS JASPEEET.
STUDENT OF NIT JALANDHAR
```